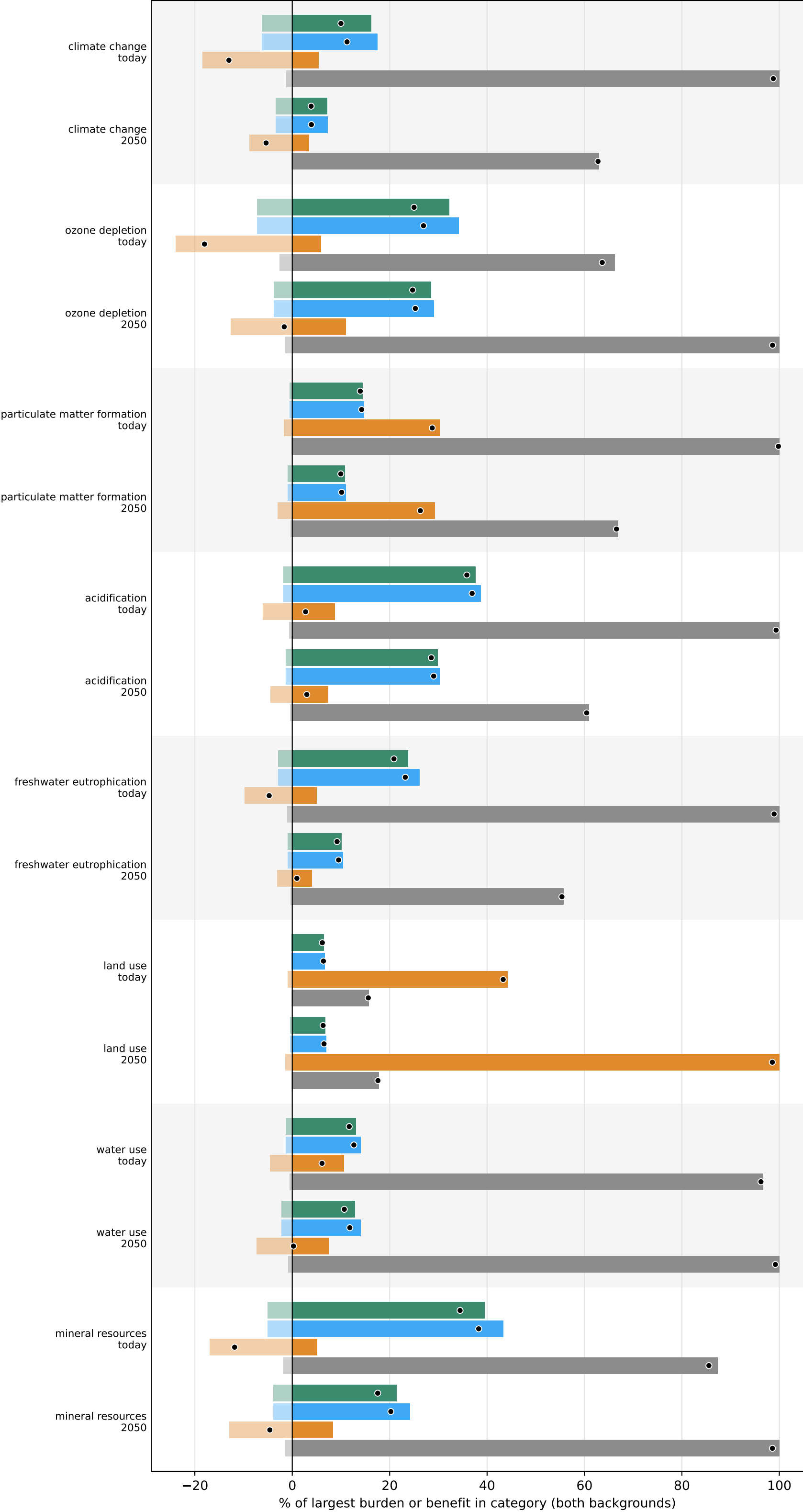


RAW coreless filament winding (beam): today vs. 2050  
beam span = 6 m, reference service life 50 yr



RAW coreless filament winding (beam)  
RAW coreless filament winding (beam) - scaled up  
glulam beam  
reinforced concrete beam  
benefits (lighter, left)  
net impact